Exp No:20. Experiment to understand pandas library use cases in Data science.

Code:

```
import numpy as np

import pandas as pd


# Creating DataFrame from list[span_3](start_span)[span_3](end_span)

data_list = [[1, 'Smith', 50000], [2, 'John', 60000]]

df = pd.DataFrame(data_list)

df.columns = ['Empd', 'Name', 'Salary']


print(df.info())

print(df.head())

print(df.tail())


# Reading from CSV file[span_4](start_span)[span_4](end_span)

df = pd.read_csv('/content/employee.csv')


print(df.head())

print(df.tail())

print(df.info())


print(df['Salary'])
```

```
print(df['Salary'].var())[span_5](start_span)[span_5](end_span)
print(df['Salary'].std())[span_6](start_span)[span_6](end_span)
print(df.describe())[span_7](start_span)[span_7](end_span)
print(df.describe(include='all'))[span_8](start_span)[span_8](end_span)

emparray = df.values[span_9](start_span)[span_9](end_span)
employee_df = pd.DataFrame(emparray, columns=df.columns)[span_10](start_span)[span_10](end_span)
print(employee_df)[span_11](start_span)[span_11](end_span)
```

Output:

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

	R&D Spend	Administration	Marketing Spend	State	Profit
45	1000.23	124153.04	1903.93	New York	64926.08
46	1315.46	115816.21	297114.46	Florida	49490.75
47	0.00	135426.92	0.00	California	42559.73
48	542.05	51743.15	0.00	New York	35673.41

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000

	emp id	name	salary
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000

	salary
0	5000
1	6000
2	7000
3	5000
4	8000
5	3000
6	6000

	emp id	name	salary
count	7.000000	7	7.000000
unique	NaN	6	NaN
top	NaN	SRI HARSHAVARDHANAN R	NaN
freq	NaN	2	NaN
mean	4.000000	NaN	5714.285714
std	2.160247	NaN	1603.567451
min	1.000000	NaN	3000.000000
25%	2.500000	NaN	5000.000000
50%	4.000000	NaN	6000.000000
75%	5.500000	NaN	6500.000000

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

Result:

Thus the pandas library use cases in data science was analysed and this code executed successfully.